DIPLOMA PROGRAMME PRACTICE · SL · SET A

Chemistry / Paper 1A

Question booklet · 30 marks · 90 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator and a clean Chemistry data booklet. Record choices online or on separate paper.

QUESTION 1 [1]

A mixture contains sand, a water-soluble salt and magnetic iron grains. Water is added and the mixture filtered. Which procedure recovers all three components separately?

A Apply a magnet to the filtrate and evaporate the residue
B Evaporate the filtrate and use a magnet on the dried residue
C Evaporate the whole mixture before separating the salt with a magnet
D Distil the filtrate to collect the salt as distillate

QUESTION 2 [1]

An element has isotopes of relative masses 63 and 65. An enriched sample has relative atomic mass 64.30. What fraction of its atoms have 36 neutrons if the atomic number is 29?

A 0.700
B 0.350
C 0.300
D 0.650

QUESTION 3 [1]

Species X has 17 protons and the ground-state configuration of argon. Which change produces neutral X without changing its nucleus?

A Add one electron to 4s
B Remove one electron from 3p
C Remove one electron from 3s
D Add one proton to the nucleus

QUESTION 4 [1]

A 0.880 g compound containing only C, H and O yields 1.760 g CO_2 and 0.720 g H_2O on complete combustion. Use C=12.0, H=1.00, O=16.0. What is its empirical formula?

A CH_2O
B C_4H_8O
C C_2H_6O
D C_2H_4O

QUESTION 5 [1]

A hydrated salt initially weighs 3.60 g. After heating, the apparent anhydrous mass is 2.16 g. Some water actually remains. Relative to the true hydration number, the value calculated from these masses is what?

A Too small because water loss is underestimated and salt amount overestimated

B Too large because both measured masses are too large

C Too large because the residue is heavier than dry salt

D Unchanged because both errors cancel

QUESTION 6 [1]

A fixed amount of ideal gas occupies 480 cm³ at 300 K and 96.0 kPa. It is compressed to 320 cm³ and cooled to 270 K. What is the final pressure?

A 57.6 kPa

B 160.0 kPa

C 86.4 kPa

D 129.6 kPa

QUESTION 7 [1]

A crystal conducts only after melting. It contains X^{3+} and Y^{2-}, and its formula unit contains no other ions. Which formula and explanation of molten conductivity fit?

A X_2Y_3; mobile ions carry charge

B X_2Y_3; neutral formula units carry charge

C XY; the charges cancel in pairs

D X_3Y_2; electrons are released on melting

QUESTION 8 [1]

At the same pressure, HF boils at a higher temperature than HCl despite its smaller molar mass. Which change absorbs most of the energy when HF boils?

A Breaking the H–F covalent bond in each molecule

B Separating pre-existing H^+ and F^- ions in the liquid

C Removing an electron from fluorine

D Overcoming intermolecular attractions including hydrogen bonds

QUESTION 9 [1]

Which pair has the same number of electron domains around each central atom but different molecular geometries?

A CO_2 and BF_3

B BF_3 and NH_3

C CH_4 and CO_2

D NH_3 and H_2O

QUESTION 10 [1]

A metal wire and its dilute substitutional alloy both conduct. The alloy resists bending more strongly. Which model accounts for both observations?

A Mobile electrons persist, while size differences hinder layers slipping

B Alloying removes the positive ions from the lattice

C Every alloy atom forms four localized bonds

D Ions in the alloy move freely through the solid

QUESTION 11 [1]

A segment of an addition polymer is –CH_2–$CH(CH_3)$–CH_2–$CH(CH_3)$–. Which statement follows from the connectivity?

A The monomer is ethene and methanol is eliminated

B The monomer is propene and each repeat retains all its atoms

C The polymer contains one double bond per repeat

D The monomer is propane and H_2 is eliminated per repeat

QUESTION 12 [1]

Four species have ten electrons. Which has both the largest nuclear charge and the smallest ionic radius?

A F^-

B Al^{3+}

C Na^+

D O^{2-}

QUESTION 13 [1]

In $S_2O_3^{2-}$ the two S sites are inequivalent. Which deduction is justified by the formula alone, taking oxygen as −2?

A Each sulfur must be +2

B One sulfur must be +6 and the other −2

C The mean oxidation state of sulfur is +2

D The total oxidation number of the sulfur atoms is −4

QUESTION 14 [1]

Which pair consists of structural isomers that belong to different homologous series?

A Butan-1-ol and butan-2-ol

B Ethanoic acid and ethanol

C Propan-1-ol and methoxyethane

D Propene and propane

QUESTION 15 [1]

Which condensed formula has an OH-bearing carbon bonded directly to two other carbon atoms?

A $(CH_3)_2CHCH_2OH$

B $CH_3CH(OH)CH_2CH_3$

C $(CH_3)_3COH$

D $CH_3CH_2CH_2CH_2OH$

QUESTION 16 [1]

Two 40.0 cm³ solutions, initially at the same temperature, react to give a temperature rise of 3.00 K. Assume density 1.00 g cm^{-3} and c=4.18 J g^{-1} K^{-1}. If 0.0100 mol reacts, what is the molar enthalpy change?

A −100 kJ mol^{-1}

B −50.2 kJ mol^{-1}

C +100 kJ mol^{-1}

D −1.00 kJ mol^{-1}

QUESTION 17 [1]

Given X→Y, ΔH=−42 kJ mol^{-1}, and Z→Y, ΔH=+17 kJ mol^{-1}, what is ΔH for X→Z on the same molar basis?

A −59 kJ mol^{-1}

B +59 kJ mol^{-1}

C +25 kJ mol^{-1}

D −25 kJ mol^{-1}

QUESTION 18 [1]

A burner takes in 18.0 g of fuel with specific energy 30.0 kJ g^{-1} and delivers 324 kJ to water. How much energy per gram of fuel escapes this useful transfer?

A 30.0 kJ g^{-1}

B 12.0 kJ g^{-1}

C 18.0 kJ g^{-1}

D 21.6 kJ g^{-1}

QUESTION 19 [1]

A sealed vessel contains 0.250 mol propane and 0.900 mol O_2. For complete combustion to CO_2 and H_2O, what is the greatest amount of CO_2 possible?

A 0.180 mol

B 0.540 mol

C 0.750 mol

D 0.900 mol

QUESTION 20 [1]

A 2.00 g impure carbonate gives 0.336 dm^3 CO_2 at a molar gas volume of 24.0 dm^3 mol^{-1}. Only $CaCO_3$ reacts and acid is in excess. Use $M(CaCO_3)=100$ g mol^{-1}. What is the mass percentage $CaCO_3$?

A 84.0%
B 140%
C 70.0%
D 35.0%

QUESTION 21 [1]

A 20.00 cm^3 aliquot of diprotic acid requires 18.00 cm^3 of 0.1200 mol dm^{-3} NaOH for complete neutralization. What is the original acid concentration?

A 0.2160 mol dm^{-3}
B 0.06667 mol dm^{-3}
C 0.05400 mol dm^{-3}
D 0.1080 mol dm^{-3}

QUESTION 22 [1]

Identical excess acid is added to equal masses of pure carbonate as powder or chips. Which pair describes the powder relative to the chips if all CO_2 is collected at the same T and p?

A Same initial slope, smaller final volume
B Greater initial slope, greater final volume
C Greater initial slope, same final volume
D Same initial slope, greater final volume

QUESTION 23 [1]

A reaction gives 22.0 cm^3 of gas at 15.0 s, 38.0 cm^3 at 35.0 s and 50.0 cm^3 at 55.0 s. What is the average gas production rate from 15.0 to 55.0 s?

A 0.700 cm^3 s^{-1}
B 0.600 cm^3 s^{-1}
C 0.909 cm^3 s^{-1}
D 0.800 cm^3 s^{-1}

QUESTION 24 [1]

At fixed temperature a gaseous equilibrium $2NO_2 \rightleftharpoons N_2O_4$ is compressed. Which combination describes the new equilibrium?

A More NO_2 in moles; K increases
B More N_2O_4 in moles; K unchanged
C More N_2O_4 in moles; K increases
D No change in composition; K decreases

QUESTION 25 [1]

At equilibrium for $H_2+I_2 \rightleftharpoons 2HI$ in one gas phase, concentrations in mol dm^{-3} are 0.20, 0.10 and 0.60 respectively. What is Kc using the usual concentration expression?

A 30
B 0.0556
C 3.0
D 18

QUESTION 26 [1]

At 298 K, 10.0 cm^3 of 0.100 mol dm^{-3} HCl is mixed with 30.0 cm^3 of water. Assume complete dissociation and additive volumes. What is the pH?

A 2.00
B 1.40
C 1.60
D 0.60

QUESTION 27 [1]

A galvanic cell operates by $Mg(s)+Cu^{2+}(aq) \rightarrow Mg^{2+}(aq)+Cu(s)$. Which simultaneous changes occur?

A Mg electrode loses mass and electrons flow externally toward Cu
B Mg^{2+} is reduced and the Mg electrode grows
C Cu^{2+} is oxidized and anions travel through the wire
D Cu electrode loses mass and electrons flow toward Mg

QUESTION 28 [1]

Molten $MgCl_2$ is electrolysed with inert electrodes. For every 0.030 mol Mg formed, how much Cl_2 forms and at which electrode?

A 0.015 mol at the cathode
B 0.030 mol at the cathode
C 0.060 mol at the anode
D 0.030 mol at the anode

QUESTION 29 [1]

Which step both consumes a radical and produces a new radical in chlorination of ethane?

A $C_2H_5\bullet + Cl\bullet \rightarrow C_2H_5Cl$
B $2C_2H_5\bullet \rightarrow C_4H_{10}$
C $Cl\bullet + C_2H_6 \rightarrow HCl + C_2H_5\bullet$
D $Cl_2 \rightarrow 2Cl\bullet$

QUESTION 30 **[1]**

When OH^- reacts with bromoethane to form ethanol, which statement identifies the electron-pair donor and the displaced species?

A OH^- donates a lone pair; Br^- leaves

B Br^- donates a lone pair; OH^- leaves

C OH^- accepts a pair; Br^+ leaves

D The carbon donates a lone pair; Br• leaves